

SMART TECHNOLOGY REPORT

Artificial Intelligence and Computer Vision | Research Intelligence Overview | 2023

Executive Summary

The Artificial Intelligence and Computer Vision domain produced 89 total research outputs in 2023, including 86 publications, 3 patents, 0 grants and 0 theses. The year-on-year growth was -9.18%, while the long-term CAGR stands at 14.32%. Across all participating institutions, the average Grant Impact Score was 31.86.

Year Overview

2023 Snapshot

The selected technology recorded **89** total research outputs during the selected year. Publications contributed **86** outputs, while patents contributed **3**.

Grant and thesis activity recorded **0** and **0** outputs respectively.

86	3	0	0
PUBLICATIONS	PATENTS	GRANTS	THESES
TOTAL RESEARCH OUTPUT			89
YEAR-ON-YEAR GROWTH		COMPOUND ANNUAL GROWTH RATE	
-9.18%		14.32%	

Understanding the Growth Metrics

YoY Growth: Measures how much research output changed compared with the previous year. A positive value means output increased, while a negative value means output declined.

CAGR: Measures the average annual growth rate over the entire analysed period. Unlike YoY, which focuses on one year-to-year change, CAGR provides a longer-term view of the research trend.

Grant Impact

The average Grant Impact Score for the selected technology was **31.86**. This provides an additional indicator for understanding the impact associated with grant-supported research activity.

5. Research Trend & Year Analysis

Research activity for **Artificial Intelligence and Computer Vision** across the analysed period, with emphasis on the selected year **2023**.

Research Output Trend

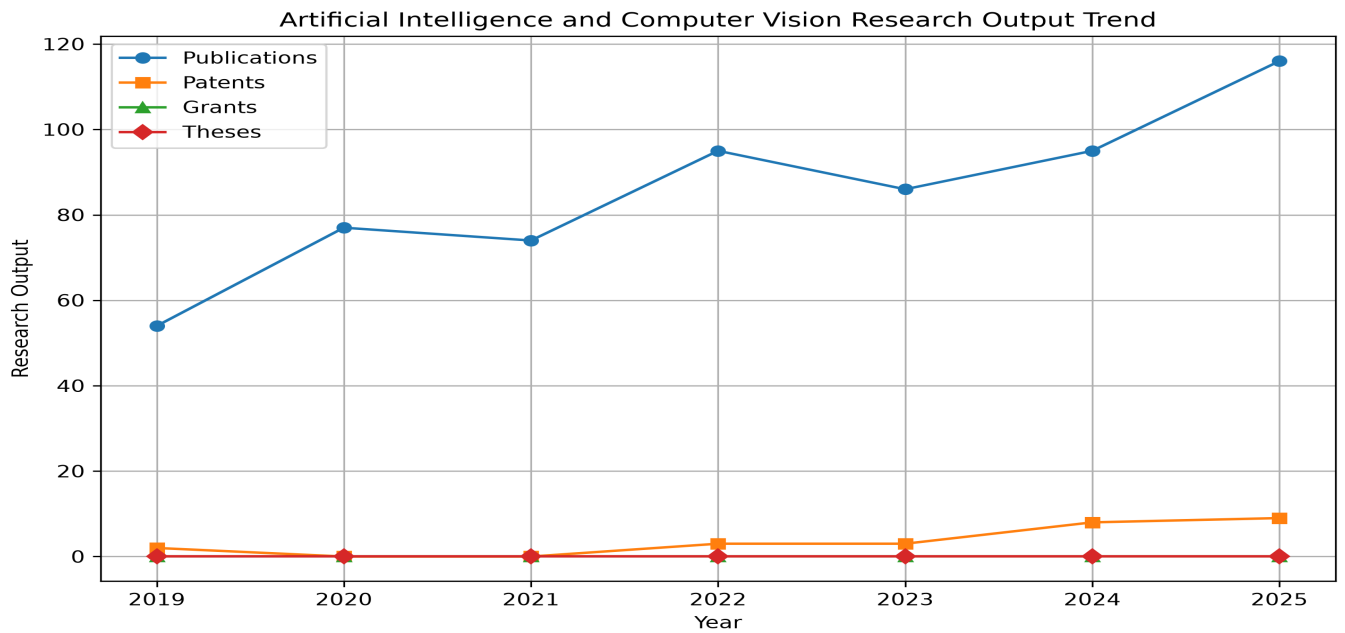


Figure 1. Year-wise research output trend for Artificial Intelligence and Computer Vision. The graph compares publications, patents, grants and theses.

Trend Interpretation

Short-Term View Research output decreased by 9.18% compared with the previous year. This indicates a short-term decline in output.	Long-Term View Over the longer period, the domain recorded a positive CAGR of 14.32% . This indicates that despite year-to-year fluctuations, research activity has grown overall.
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2023 Research Overview

In **2023**, the **Artificial Intelligence and Computer Vision** domain recorded **89** total research outputs. This included **86** publications, **3** patents, **0** grants and **0** theses.

Funding Observation

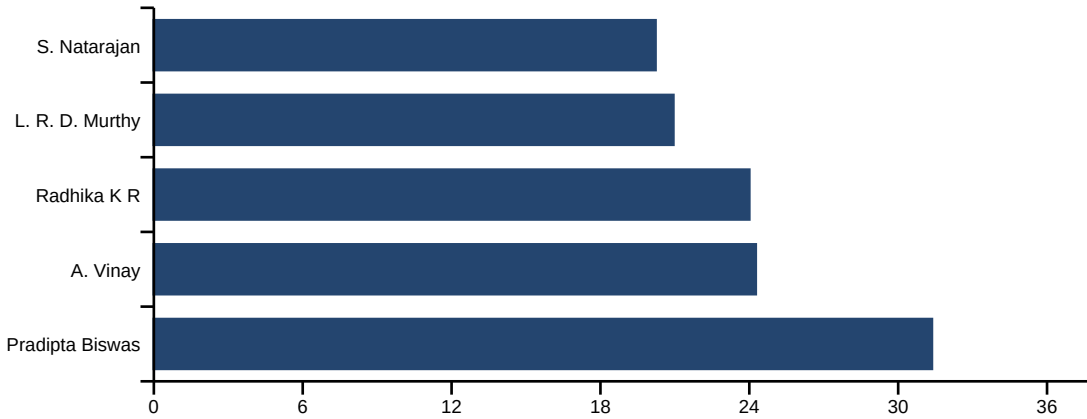
The selected year contains no grant-related outputs in the current dataset. Therefore, grant activity does not contribute to the research-output total for 2023. The Grant Impact Score is retained as a supplementary funding-related indicator.

6. Research Landscape

This section identifies the leading researchers and institutions associated with **Artificial Intelligence and Computer Vision**.

Top Researchers

Top Researchers by Research Score



Rank	Researcher	Institution	Score
	Pradipta Biswas	Indian Institute of Science Bangalore	31.36
	A. Vinay	PES University	24.26
	Radhika K R	B.M.S. College of Engineering	24.00
	L. R. D. Murthy	Indian Institute of Science Bangalore	20.94
	S. Natarajan	PES University	20.22

7. Leading Institutions

Institutions are ranked according to their research output within the selected technology.

Institution output values are currently zero or unavailable for the selected technology/year, so a meaningful institution comparison chart cannot be displayed.

Rank	Institution	Research Output
	Indian Institute of Science Bangalore	0
	R V College of Engineering	0
	CMR Institute of Technology	0
	B.M.S. College of Engineering	0
	International Institute of Information Technology Bangalore	0
	Dayananda Sagar College of Engineering	0
	M.S. Ramaiah Institute of Technology	0

8. Key Insights

- Research output decreased by 9.18% compared with the previous year.
- The positive CAGR of 14.32% indicates long-term growth across the analysed period.
- **Pradipta Biswas** ranked first with a research score of **31.36**.
- The selected year recorded **89** total research outputs.

9. Recommendations

- Research output has declined. Strategic funding and collaboration should be encouraged.
- Grant funding should be strengthened to improve technology development.

SMART Report Note: This report is automatically generated from the research datasets and analytical outputs processed by the SMART trend engine.